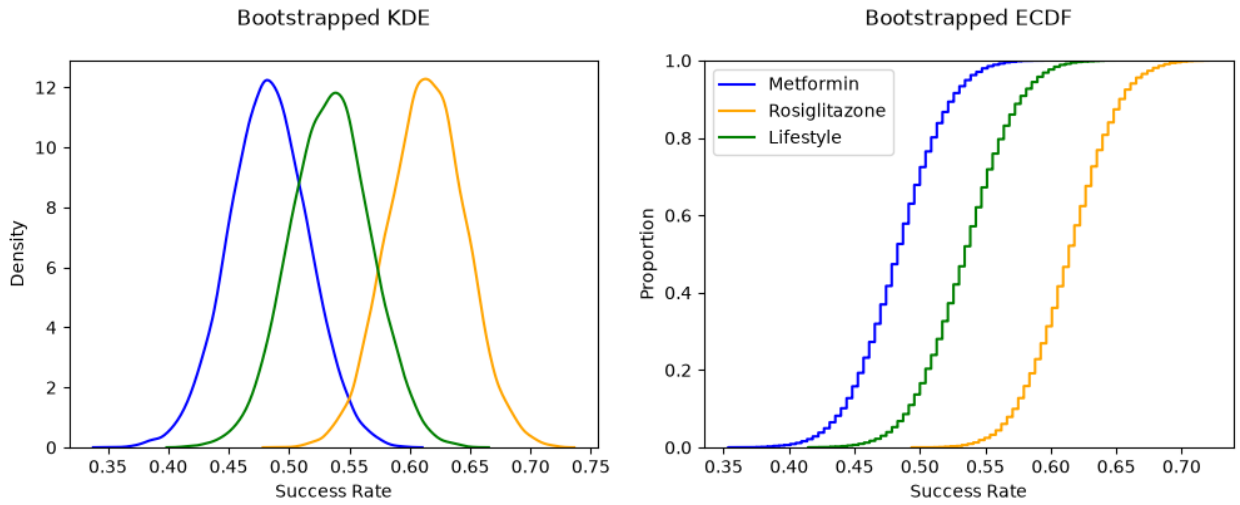
Report: Clinical Trial for Pre-Diabetes Treatment

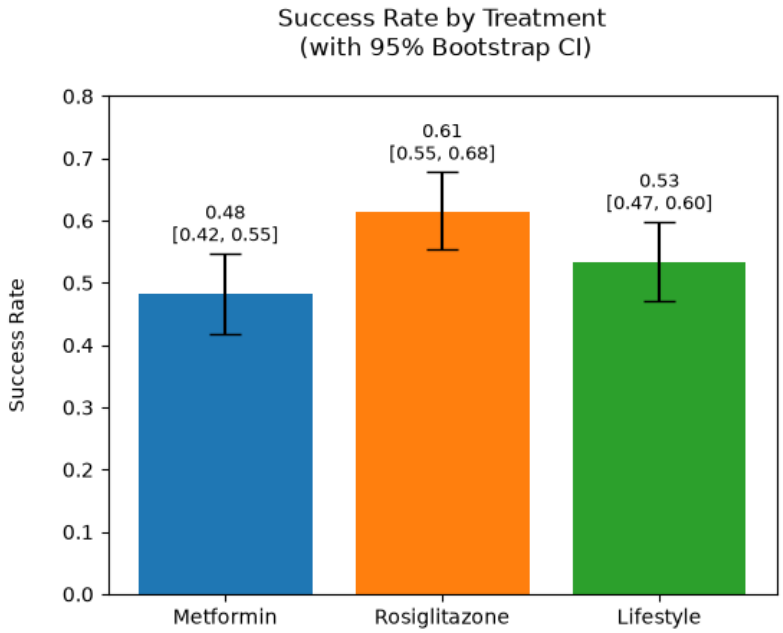
The goal of this randomized controlled clinical trial is to consider three different treatments for pre-diabetes in terms of how they help slow the progression to full diabetes. The trial involved randomizing pre-diabetic participants to one of three interventions: Metformin (n = 232), Rosiglitazone (n = 233), or Lifestyle adjustments of diet and exercise (n = 234). For each treatment group, outcomes were recorded based on success (coded as 1s) and failure (coded as 0s), where a failure constitutes a patient developing full diabetes. No control treatment (no treatment) is included, so we compare interventions to each other.

Treatment	N	Successes	Failures
Rosiglitazone	233	143	90
Lifestyle	234	125	109
Metformin	232	112	120

To do this, we investigated the treatment effect for each pair of treatments, the average treatment effect (ATE), which is the difference in success rates. Then, to determine the uncertainty surrounding these estimates, we resampled with replacement 10,000 times within each of the three treatment groups. We then constructed 90%, 95%, and 99% confidence intervals for each pairwise ATE.



The figure above visualizes the bootstrapped distributions for success rates for each treatment. The KDE shows overlap between the Lifestyle intervention and the other two treatments, while a clearer separation between Metformin and Rosiglitazone is shown.



The bar chart above shows the success rates for each treatment at the 95% confidence level. Rosiglitazone had the highest observed success rate (0.61), while metformin had the lowest (0.48). The only significant difference that held across all three confidence levels was between Metformin and Rosiglitazone. The ATE is -0.131, meaning that the difference favors Rosiglitazone, and the confidence interval excludes zero at all three levels. Thus, this difference between Rosiglitazone and Metformin performance is statistically significant, with Rosiglitazone having a higher success rate than Metformin. The only other significant difference came at the 90% confidence level when comparing Rosiglitazone and Lifestyle, with an ATE of 0.08, in favor of Rosiglitazone. The 95% and 99% intervals contain zero, which indicates that the evidence that Rosiglitazone is significantly different compared to the success rate of Lifestyle is not entirely conclusive. At none of the three confidence levels was a statistically significant difference between Metformin and Lifestyle detected. Thus, there is not enough evidence here to distinguish between these two treatments and their respective success rates. Confidence intervals for the ATEs at all three confidence levels are shown in the table below.

Comparison	ATE	90% CI	95% CI	99% CI
Met – Rosi	-0.131	**[-0.204, -0.058]**	**[-0.221, -0.041]**	**[-0.247, -0.019]**
Met – Lifestyle	-0.051	[-0.128, 0.022]	[-0.142, 0.038]	[-0.172, 0.069]
Rosi – Lifestyle	0.08	**[0.007, 0.152]**	[-0.006, 0.169]	[-0.040, 0.199]
***Bolded where significant**				

The strongest evidence supports Rosiglitazone as being the leading performer of the three treatments. It outperforms Metformin and the evidence is indicative of a potential

advantage over the Lifestyle intervention, while not fully conclusive. Metformin has the weakest evidence, being significantly worse than Rosiglitazone and not distinguishable from Lifestyle.